

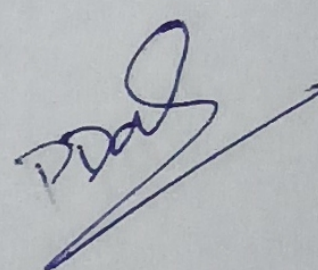
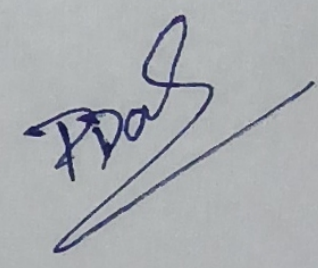
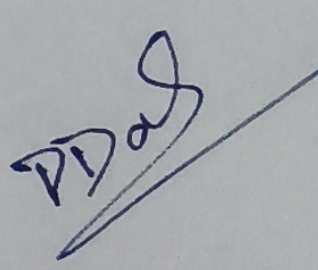
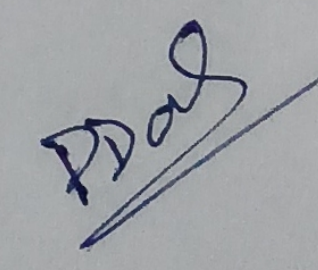
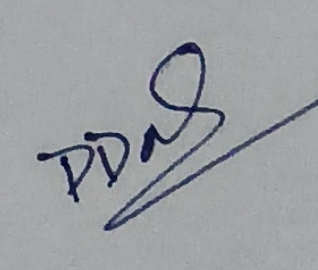
Department of Computer Science & Engineering
 Global Campus, Jakkasandra Post, Kanakapura Taluk, Ramanagara District, Pin Code: 562 112

FACULTY OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

PCL Attendance

Team No. : COEPCL-0213.

Project Title : ZK Credential Verification with Blockchain and LLM parsing.

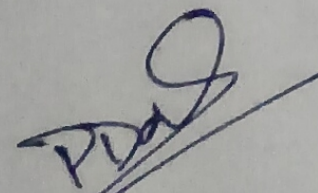
Sl. No.	Date of the Meeting	Points Discussed	Members Absent (Roll No)	Signature of the Mentor
1.	16/1/26	Project understanding & Technology Research; learning about Zero Knowledge proofs, Circom, Fast API, React, Docker and Blockchain integration.	-	
2.	24/1/26	Configured the development environment using Docker compose and Setup postgresql, Redis, backend, and frontend Services Successfully	-	
3.	31/1/26	Developed backend APIs using FastAPI for document upload, Job tracking, and Secure JWT-based authentication.	-	
4.	6/1/26	Implemented Secure document upload with multipart handling and AES-256 encryption for protected file Storage.	-	
5.	14/2/26	Integrated OCR and text extraction using Tesseract OCR and PyMuPDF for processing image and PDF documents.	-	

6.	21/2/26	Integrated the LLM to extract structured claims from documents and generate JSON-based credential information.	23BTRCTD91	<u>PDa8</u>
7.	28/2/26	Validated extracted claims using Pydantic models and generated witness data for zero knowledge proof creation.	-	<u>PDa8</u>
8.	5/3/26	Designed and compiled zero knowledge circuits in circom for age verification and commitment checking.	23BTRCT084	<u>PDa8</u>
9.	13/3/26	Generated and verified zero knowledge proofs using snarkjs and tested proof verification successfully.	-	<u>PDa8</u>
10.	27/3/26	Developed and deployed a solidity verifier smart contract using Hardhat on a test blockchain network	-	<u>PDa8</u>
11.	2/04/26	Built the React frontend with Tailwind CSS and integrated file upload, wallet connection, and status tracking features.	23BTRCT084	<u>PDa8</u>
12.	9/04/26	Performed end-to-end testing, fixed bugs, improved performance, and completed final documentation and presentation materials.	-	<u>PDa8</u>

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PROJECT CENTRIC LEARNING (PCL) PHASE 2

PROJECT TITLE		PROJECT TEAM NO.		CoEPCL-0213 [41]
		ZK Credential Verification with Blockchain and LLM Parsing		
TEAM MEMBERS				
S. No.	USN No.	NAME OF THE STUDENT	CLASS	MOBILE NO.
1	23BTRCTD72	M. G. Prasanna Sai	CSE* B	7207184364
2	23BTRCTD84	K. Mokshitua	CSE* B	8919838978
3	23BTRCTD71	M. Charan Kumar	CSE* B	8885392756
4	23BTRCTD91	N. Ukesh Reddy	CSE* C	7337367536
5	23BTRCTD51	K V Uddip Ganesh	CSE* A	7995951956
6	23BTRCTD84	Naaya S	CSE* C	9003509371
DETAILS OF THE PROJECT GUIDE				
NAME		Prof. Pranjatika Das		
DESIGNATION		Professor		
DEPARTMENT		Computer Science & Engineering (General)		
eMail ID		pranjatika.das@jaiuniversity.ac.in		
Phone No.		736559796		
DOMAIN OF THE PROJECT				
The domain of the project include Cyber Security, Blockchain Technology, Artificial Intelligence and (ZKP) Zero - knowledge proof systems.				
Scope of the Project				
1. Secure User Authentication and Encrypted document Management				
2. AI - Based OCR, claim Extraction, and Validation.				
3. Zero - knowledge Proof Generation with Blockchain Verification				
4. Scalable Backend Processing and Frontend Dashboard Deployment.				


PROJECT GUIDE

PROJECT COORDINATOR

Existing System:-

The existing credential verification systems are mainly document-centric and disclosure-heavy systems. Users are required to upload full documents for verification purpose.

Features:-

- * Manual or semi-automatic document verification.
- * Full document sharing
- * Centralized storage of sensitive data
- * Rule based extraction systems

Drawbacks of Existing Systems:-

- * Exposure of personal information.
- * Risk of data breaches
- * Unauthorized access.
- * High dependency on centralized trust
- * Limited privacy protection
- * High storage and compliance burden

Problem Identification

The problems identified are:-

1. Privacy Leakage
 2. Data breach risks
 3. Trust Fragmentation
 4. Document Heterogeneity
- 5. Lack of Cryptographic Assurance
 - 6. Compliance Challenges.

Proposed System:-

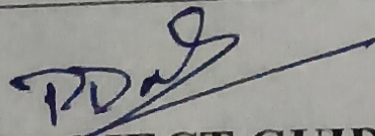
The proposed system is a zk Credential Verification system using blockchain and LLM parsing that enables privacy-preserving credential verification using AI and cryptography.

Features of Proposed System:-

1. LLM based document parsing.
2. OCR based processing
3. Zero Knowledge proof generation
4. Blockchain Verification
5. Minimal Data Disclosure
6. Secure Storage
7. Asynchronous scalable processing
8. Trustless verification

No. of Modules

Sl. No.	Name of the Module	Remarks
1.	User Authentication and Security	
2.	Document Upload and storage Module	
3.	OCR and Document processing module.	
4.	LLM - Based claim Extraction Module.	
5.	Claim validation and Normalization Module	
6.	Blockchain and smart contract module.	


PROJECT GUIDE

PROJECT COORDINATOR